

Correct or Round? Granulometries by Digital Disks

Anonymous Author(s)

Anonymous Institution

Abstract. Granulometries by disks are the standard morphological tool for measuring size distributions. On $\mathbb{Z}^2$ they face a conflict: the structuring elements should look like Euclidean disks, and they should satisfy Matheron’s absorption axiom, i.e. each disk must be open with respect to every smaller one. We first show that the digital disks of common libraries violate the axiom for most pairs of radii, so that size distributions computed with them can increase with the radius. Families that satisfy the axiom are classically built as Minkowski chains of small structuring elements, such as neighbourhood sequences, periodic lines or Bresenham decompositions, and they are visibly polygonal. We prove that this is unavoidable. If the mean radius of a Minkowski chain grows by at most δ per element, the Hausdorff distance between its elements and Euclidean disks of radius ρ is at least $\rho/(32\pi^2\delta^2)$ asymptotically. Hence fine granulometric chains are linearly anisotropic. An explicit chain of periodic lines attains this rate up to a constant factor of 8, and letting δ grow like $\rho^{1/4}$ yields error $\Theta(\sqrt{\rho})$. The lower bound extends to $\mathbb{Z}^n$, and for a fixed library its constant is computed exactly by linear programming and is attained by octagons. Experiments on synthetic and real images illustrate both the violation of the axiom by digital disks and the anisotropy of granulometric chains.

Keywords: Granulometry · Digital disk · Minkowski sum · Lattice polygon · Width function · Pattern spectrum

1 Introduction

Granulometries, introduced by Matheron [8], formalise sieving: a family of openings $(\gamma_\lambda)_{\lambda \geq 0}$ that are decreasing in λ and satisfy the absorption law $\gamma_\lambda \gamma_\mu = \gamma_\mu \gamma_\lambda = \gamma_{\max(\lambda, \mu)}$. The size distribution and pattern spectrum derived from them [7, 11, 12] are among the most widely used morphological descriptors. For openings by structuring elements B_λ , absorption holds if and only if B_μ is B_λ -open, i.e. a union of translates of B_λ , for all $\lambda \leq \mu$. Euclidean disks satisfy this condition, and they form the canonical isotropic granulometry.

On the grid $\mathbb{Z}^2$, practitioners face a choice that is rarely made explicit. The digital disks $\{p \in \mathbb{Z}^2 : \|p\| \leq r\}$ of common software are round, but they are *not* open with respect to each other (Sect. 3). With `scikit-image` [15], 73% of the pairs of radii up to 40 violate the condition, and size distributions computed with these disks increase with the radius on most random test images. The

classical remedy [1,6,12,14] builds the structuring elements as *Minkowski chains* $D_k = D_{k-1} \oplus C_k$, which are automatically granulometric. Their elementary sets C_k come from small libraries: the 4- and 8-neighbourhoods give octagons (neighbourhood sequences [9,3]), and there are also periodic lines [6] and Bresenham segments [1]. These families are correct but polygonal.

We show that this trade-off is a theorem, and we determine its rate.

- **Lower bound** (Theorem 1, Corollaries 1–3). If the mean radius of a Minkowski chain grows by at most δ per element, its error ε , the Hausdorff distance to Euclidean disks, satisfies $\varepsilon \geq (1 - o(1)) \rho / (32\pi^2 \delta^2)$ at radius ρ . Every fine chain is therefore *linearly* anisotropic. If instead the fitted radius grows by bounded steps, then $\varepsilon \geq c \rho^{1/3}$. For a fixed finite library the slope is an explicit linear program, and it is attained by octagons.
- **Matching upper bound** (Theorem 3). An explicit chain of periodic lines along the boundary of the square $[-m, m]^2$ has steps $\delta \leq \sqrt{2}m/\pi$ and error $\varepsilon \leq \rho/(8m^2) + 2m^2$. The optimal error is thus $\Theta(\rho/\delta^2)$ for $\rho \gtrsim \delta^4$, with constants within a factor of 8 asymptotically, and $\Theta(\sqrt{\rho})$ when $\delta \sim \rho^{1/4}$.
- **Higher dimensions** (Theorem 2). In $\mathbb{Z}^n$, $\varepsilon \geq c_n \rho \delta^{-2n}$: bounded steps force linear anisotropy in every dimension.
- **Evidence**. We quantify the violation of the axiom by standard digital disks (Table 1), compare practical families (Table 3), and test size distributions on rotated synthetic and real images (Table 4).

The message for practice is that one can be correct (granulometric) or round, but not both at all scales, and our bounds quantify the price. Related work on approximating circles by neighbourhood sequences [3] or spheres by zonohedra [5] optimises roundness for a fixed library, and Soille and Talbot [13] noted that radial decompositions by Bresenham lines do not generate granulometries. To our knowledge, no lower bound of this kind has been published.

2 Preliminaries

For a finite set $X \subset \mathbb{Z}^n$ and a unit vector u , let $h_X(u) = \max_{p \in X} \langle p, u \rangle$ be the support function and $w_X(u) = h_X(u) + h_X(-u)$ the *width function*, i.e. the width of $\text{conv } X$ in direction u . Both are additive under Minkowski addition [10]:

$$h_{X \oplus Y} = h_X + h_Y, \quad w_{X \oplus Y} = w_X + w_Y. \quad (1)$$

The *anisotropy* of X is $A(X) = \max_u w_X - \min_u w_X$, and its *mean radius* is $\bar{r}(X) = \frac{1}{2} \int w_X d\sigma$, where σ is the uniform probability measure on the sphere; $\bar{r}$ is additive as well. We say that X is ε -*round with radius* ρ if $d_H(\text{conv } X, B(c, \rho)) \leq \varepsilon$ for some centre c . Then $|h_X(u) - \langle c, u \rangle - \rho| \leq \varepsilon$, and hence

$$2\rho - 2\varepsilon \leq w_X \leq 2\rho + 2\varepsilon, \quad A(X) \leq 4\varepsilon, \quad |\bar{r}(X) - \rho| \leq \varepsilon. \quad (2)$$

The opening of Y by a structuring element D is $Y \circ D = \bigcup \{D + t : D + t \subseteq Y\}$. A family (D_k) generates a granulometry if and only if $D_l \circ D_k = D_l$ for

Table 1. Violations of the absorption condition by standard digital disks, $1 \leq s < r \leq 40$.

family	violating pairs (s, r)	violating pairs $(r - 1, r)$	images with non- monotone size distr.
<code>scikit-image disk</code>	571/780 (73%)	30/39	253/400
OpenCV <code>MORPH_ELLIPSE</code>	628/780 (81%)	36/39	241/400

all $k \leq l$ [8,11]; since the relation is transitive, consecutive elements suffice. A *Minkowski chain* is a family with $D_k = D_{k-1} \oplus C_k$, and every Minkowski chain is granulometric. Its *increments* are the C_k , its *steps* are $\bar{r}(D_k) - \bar{r}(D_{k-1}) = \bar{r}(C_k)$, and, if all C_k belong to a finite set $\mathcal{L}$ (up to translation), $\mathcal{L}$ is its *library*.

3 Digital Disks Are Not Granulometric

We tested the condition “ D_r is D_s -open” for all pairs $1 \leq s < r \leq 40$, for the digital disks of `scikit-image` [15] (`disk`, the Gauss digitisation $\{p : \|p\| \leq r\}$) and of OpenCV [2] (`MORPH_ELLIPSE`). As Table 1 shows, most pairs violate it, including most consecutive pairs. When D_r is not D_s -open, $X \circ D_r \not\subseteq X \circ D_s$ for some X , and the size distribution $r \mapsto |X \circ D_r|$ can *increase*, so the pattern spectrum takes negative values. On random 120×120 images made of 3 to 11 disks, rectangles or digital disks ($r \leq 15$), this happens for about 60% of the images. For example, in one image the opened area grows from 2053 to 2089 pixels between $r = 8$ and $r = 9$, and `scipy.ndimage`, `scikit-image` and OpenCV agree up to border conventions.

One could try to extract a granulometric *subsequence* of digital disks. Over all origin-centred disks $\{p : \|p\|^2 \leq n\}$ with $n \leq 900$, the largest radius reachable by a chain of consecutively open disks with radius gaps at most G is 2.0 for $G = 1$, 9.9 for $G = 2$ and 21.8 for $G = 3$, and radius 30 is reached only with $G \geq 5$. Digital disks thus give at best very coarse granulometries.

4 Lower Bounds

Additivity (1) turns roundness of every element of a chain into a constraint on the increments. If consecutive elements are both nearly round, every increment has small diameter, so all its edges are short lattice vectors. Short lattice vectors leave a gap of directions around each axis, and a polygon whose edges avoid an angular interval cannot be round.

Lemma 1 (direction gap). *Let $D \geq 1$. No nonzero $(a, b) \in \mathbb{Z}^2$ with $a^2 + b^2 \leq D^2$ has a direction angle in $(0, \arctan(1/D))$.*

Proof. A direction in that interval requires $a > 0$ and $b \geq 1$, hence $b/a \geq 1/a \geq 1/D$. $\square$

Theorem 1. *Let $(D_k)_{k=0}^K$ be a Minkowski chain in $\mathbb{Z}^2$ whose elements are ε -round with radii ρ_k , and let $D \geq 1$ bound the diameters of its increments. With $\alpha = \arctan(1/D)$,*

$$8\varepsilon \geq 2(\rho_K - \rho_0 - 2\varepsilon)(1 - \cos(\alpha/2)). \quad (3)$$

Proof. Step 1. Every edge of $\text{conv } C_k$ is a lattice vector of norm at most D . *Step 2.* Let $P = \sum_k \text{conv } C_k$ and $Q = \frac{1}{2}(P - P)$. By (1), $W := w_{D_K} - w_{D_0} = w_P = 2h_Q$. The edges of the centrally symmetric polygon Q are parallel to edges of the $\text{conv } C_k$, so by Lemma 1 none has a direction in $(0, \alpha)$. Hence there is an interval I of normal angles of length at least α on which a single vertex q of Q attains the support: $h_Q(\theta) = |q| \cos(\theta - \theta_q)$ on I . *Step 3.* We may assume $\rho_K - \rho_0 > 2\varepsilon$, since otherwise (3) is trivial. Then $h_Q > 0$, so $|\theta - \theta_q| < \pi/2$ on I , and on an interval of length α a cosine varies by at least $1 - \cos(\alpha/2)$ times its amplitude. Since $|q| \geq \min h_Q = \frac{1}{2} \min W \geq \rho_K - \rho_0 - 2\varepsilon$ by (2), we get $\max W - \min W \geq 2(\rho_K - \rho_0 - 2\varepsilon)(1 - \cos(\alpha/2))$. *Step 4.* Finally $\max W - \min W \leq A(D_K) + A(D_0) \leq 8\varepsilon$ by (2). $\square$

Writing $c = 1 - \cos(\alpha/2) = \frac{1}{8D^2}(1 + O(D^{-2}))$, (3) reads $\varepsilon \geq (\rho_K - \rho_0)c/(4 + 2c)$. Different constraints on the steps give different bounds on D .

Corollary 1 (mean-radius steps). *If $\bar{r}(C_k) \leq \delta$ for all k , then $D \leq \pi\delta$ and $\varepsilon \geq (1 - o(1))(\rho_K - \rho_0)/(32\pi^2\delta^2)$ as $\delta \rightarrow \infty$. In particular, a granulometric chain with bounded steps has error growing linearly in the radius.*

Proof. By Cauchy's formula the perimeter of a planar convex set is π times its mean width, and it is at least twice its diameter. Hence $\text{diam conv } C_k \leq \frac{\pi}{2} \cdot 2\bar{r}(C_k) \leq \pi\delta$. $\square$

Corollary 2 (fitted-radius steps). *If $\rho_k - \rho_{k-1} \leq \Delta$ for all k , then $\varepsilon \geq (1 - o(1))((\rho_K - \rho_0)/512)^{1/3}$ as $\rho_K \rightarrow \infty$.*

Proof. By (1) and (2), $w_{C_k} = w_{D_k} - w_{D_{k-1}} \leq 2\Delta + 4\varepsilon$, so $D \leq 2\Delta + 4\varepsilon = 4\varepsilon(1 + o(1))$, since (3) forces $\varepsilon \rightarrow \infty$. Then (3) becomes $8\varepsilon \geq (\rho_K - \rho_0)/(64\varepsilon^2)(1 + o(1))$. $\square$

Corollary 3 (fixed library). *Let $\mathcal{L}$ be a finite library containing a set with more than one point, and let*

$$\kappa(\mathcal{L}) = \min \left\{ \frac{1}{4} A \left(\sum_C \lambda_C w_C \right) : \lambda \geq 0, \min_u \sum_C \lambda_C w_C(u) = 2 \right\}.$$

Then $\kappa > 0$, and every chain with library $\mathcal{L}$ satisfies $\varepsilon \geq (\kappa(\rho_K - \rho_0) - \frac{1}{4} A(D_0))/(1 + 2\kappa)$.

Proof. $\sum_C \lambda_C w_C$ is the width function of the polygon $\sum_C \lambda_C \text{conv } C$, which does not have constant width because bodies of constant width are strictly convex [10]; by compactness $\kappa > 0$. Write $W = w_{D_K} - w_{D_0} = \sum_C \lambda_C w_C$ and $s = \frac{1}{2} \min W \geq \rho_K - \rho_0 - 2\varepsilon$. Then W/s is feasible, so $A(W) \geq 4\kappa s$, and $A(W) \leq 4\varepsilon + A(D_0)$. $\square$

Table 2. Slope κ of Corollary 3 for classical libraries (D_4 -images included). Segment libraries contain the 8-connected digital straight segments (DSS) or the periodic lines $\{0, v\}$ of all primitive v with $\|v\| \leq m$.

library	κ	library	κ	library	κ
$\{N_4\}$ or $\{N_8\}$	0.2071	DSS, $m = \sqrt{5}$	0.0176	periodic, $m = \sqrt{5}$	0.0137
$\{N_4, N_8\}$ (octagons)	0.0412	DSS, $m = 10$	0.0028	periodic, $m = 5$	0.0038

κ is a linear program over a fine grid of directions (Table 2). For example, $\kappa(\{N_4\}) = (\sqrt{2} - 1)/2$ is the anisotropy of the square. Corollary 3 is tight for octagons: the best octagon chain has error 38.05 at $\rho = 1000$, matching the leading term $\frac{\kappa}{1+2\kappa}\rho = 38.06$ up to the $O(1)$ correction.

Higher dimensions. In $\mathbb{Z}^n$ there is no ordering of directions, but a measure argument replaces Lemma 1.

Theorem 2. *Let (D_k) be a Minkowski chain in $\mathbb{Z}^n$, $n \geq 2$, whose elements are ε -round (with respect to Euclidean balls) with radii ρ_k , and whose increments have diameter at most $D \geq 1$. There are constants $b_n, c_n > 0$ such that (3) holds with $\alpha/2$ replaced by α_n , where $\sin \alpha_n = 1/(4b_n(2D + 1)^n)$. In particular $\varepsilon \geq c_n(\rho_K - \rho_0)D^{-2n}$. If the mean-radius steps are at most δ , then $D \leq 2\delta/\mu_n$ with $\mu_n = \Gamma(\frac{n}{2})/(\sqrt{\pi} \Gamma(\frac{n+1}{2}))$, and bounded steps again force linear growth.*

Proof (sketch). The walls of the normal fan of $Q = \frac{1}{2}(P - P)$ lie in the hyperplanes $e^\perp$ for the edges e of the $\text{conv } C_k$, i.e. lattice vectors with $\|e\| \leq D$. There are at most $(2D + 1)^n$ of them, and each band $\{u : |\langle u, e/\|e\| \rangle| < s\}$ has measure at most $b_n s$. So some u_0 satisfies $|\langle u_0, e \rangle| \geq 2 \sin \alpha_n \|e\|$ for all e , and the geodesic cap of radius α_n around u_0 meets no wall. On this cap h_Q is linear, $h_Q(u) = \langle q, u \rangle$, and restricting to a great circle through u_0 and q gives a variation of at least $|q|(1 - \cos \alpha_n)$. Steps 3 and 4 are as in Theorem 1. Finally, a segment of length L has mean width $\mu_n L$. $\square$

5 A Matching Upper Bound

The lower bound is attained, up to constants, by a simple explicit chain. For $m \geq 1$, let V_m contain one vector from each pair $\pm v$ of integer points on the boundary of $[-m, m]^2$. These are $4m$ vectors of norm at most $\sqrt{2}m$, and consecutive directions differ by at most $\arctan(1/m) \leq 1/m$, the largest gap being between $(m, 0)$ and $(m, 1)$. Let Z be the polygon circumscribed to the unit disk whose edges have the directions of $\pm V_m$; it is centrally symmetric, hence $Z = \sum_{v \in V_m} [-\frac{\ell_v}{2} \hat{v}, \frac{\ell_v}{2} \hat{v}]$ with $\hat{v} = v/\|v\|$, and $1 \leq h_Z \leq \sec(\frac{1}{2m})$. For $t \geq 0$ set $n_v(t) = \lfloor t\ell_v/\|v\| \rfloor$ and

$$D(t) = \bigoplus_{v \in V_m} n_v(t) \{0, v\},$$

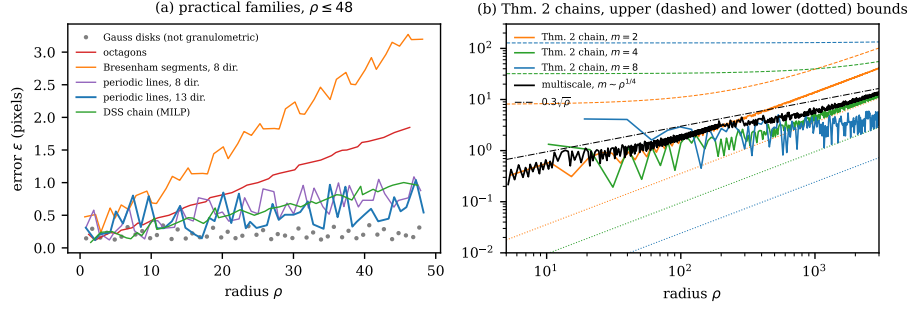


Fig. 1. (a) Error of practical families for $\rho \leq 48$ (radius steps below 2.2). (b) Chains of Theorem 3 for $m = 2, 4, 8$, with the upper bound of Theorem 3 (dashed) and the lower bound of Corollary 1 (dotted), and the multiscale chain against $0.3\sqrt{\rho}$.

the Minkowski sum of $n_v(t)$ copies of the periodic line $\{0, v\}$ [6]. As t grows, the counts increase one at a time, so the elements $D(t)$ form a Minkowski chain of periodic lines.

Theorem 3. *The chain $D(t)$ has mean-radius steps at most $\sqrt{2}m/\pi$, and every element is ε -round with radius ρ and $\varepsilon \leq \rho/(8m^2) + 2m^2$.*

Proof. A step adds one periodic line, whose mean radius is $\|v\|/\pi \leq \sqrt{2}m/\pi$. The centred support function of $\text{conv } D(t) = \sum_v n_v[0, v]$ is $H = \sum_v n_v |\langle v, u \rangle|/2 = t h_Z - \sum_v \varphi_v |\langle v, u \rangle|/2$ with $\varphi_v \in [0, 1]$. Since $\sum_v \|v\| \leq 4\sqrt{2}m^2$ and $\sec y - 1 \leq y^2$ for $y \leq \frac{1}{2}$, we get $t - 2\sqrt{2}m^2 \leq H \leq t + t/(4m^2)$. The disk of radius $\rho = (\max H + \min H)/2$ is therefore within $\varepsilon \leq t/(8m^2) + \sqrt{2}m^2 \leq \rho/(8m^2) + 2m^2$. $\square$

With $\delta = \sqrt{2}m/\pi$, Theorem 3 gives $\varepsilon \leq \rho/(4\pi^2\delta^2) + \pi^2\delta^2$, while Corollary 1 gives $\varepsilon \geq (1 - o(1))\rho/(32\pi^2\delta^2)$. For $\rho \geq 4\pi^4\delta^4$, where the constant term is dominated, the best achievable error is therefore $\Theta(\rho/\delta^2)$, with constants within a factor of 16, and of 8 asymptotically. Concatenating chains with $m_j = 2^j$, each run up to $t_j \propto 16^j$ on top of the frozen previous ones (anisotropies and radii add), yields a single chain with steps $O(\rho^{1/4})$ and error $O(\sqrt{\rho})$. By Corollary 1, $O(\sqrt{\rho})$ is optimal for chains with steps $O(\rho^{1/4})$. Numerically (Fig. 1b), the fixed- m chains have slope about $1/(18m^2)$, between the bounds $1/(64m^2)$ and $1/(8m^2)$, and the multiscale chain satisfies $\varepsilon \leq 0.28\sqrt{\rho}$ for $20 \leq \rho \leq 5 \cdot 10^4$.

Like all our bounds, Theorem 3 concerns convex hulls. Minkowski sums of periodic lines can leave isolated holes, as Jones and Soille's decompositions also do [6]. Adding the 2×2 square to the chain reduces them to 0.2–2% of the hull's lattice points in our tests, and holes do not affect the granulometric property.

Table 3. Error ε (pixels) of families for $\rho \leq 48$ with radius steps below 2.2, and the largest radius reached with $\max \varepsilon \leq 1$.

family	granulometric	max ε	mean ε	at ρ at $\varepsilon \leq 1$
Gauss disks (disk)	no	0.35	0.21	–
octagons (neighbourhood seq.)	yes	1.85	0.98	25
Bresenham/DSS, 8 directions	yes	3.27	1.75	12
periodic lines, 8 dir., LP weights	yes	1.09	0.60	41
periodic lines, 13 dir., LP weights	yes	1.04	0.48	46
DSS, $\ v\ \leq 6$, MILP	yes	1.00	0.57	44

6 Practical Families and Experiments

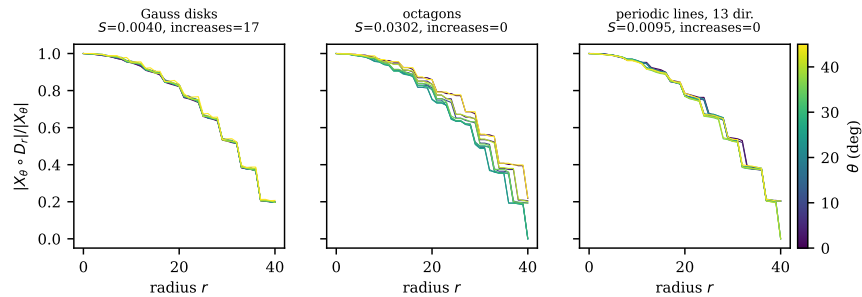
Families. Table 3 compares families with radius steps below 2.2, i.e. about one element per integer radius, for $\rho \leq 48$. For the fixed libraries (Bresenham/DSS segments as in [1], periodic lines as in [6]), we use the optimal weights λ^* of Corollary 3. We move towards the multiplicities $\text{round}(t\lambda^*)$ one elementary set at a time, choosing at each step the set that keeps the element roundest; the chains contain the 2×2 square. We also include a chain of individual DSS computed by mixed-integer programming (HiGHS [4]), which minimises the maximal error over $r \leq 48$. All elements of these granulometric families were verified to be hole-free. The optimised chains halve the error of octagons and reach about a third of that of Bresenham decompositions, but they remain three times less round than Gauss disks, in line with Sect. 4. Relaxing the step size helps, as Theorem 1 predicts through D : with radius steps up to 4.1, the 13-direction periodic chain reaches 0.41. Conversely, if multiplicities could be real, a monotone family of DSS with $\|v\| \leq 45$ would have error at most 0.43 on a grid of radii up to 2000, as we found by solving a linear program; we do not claim that this persists for all radii. Integrality is what Step 1 of the proof exploits.

Size distributions. For each scene X and angle $\theta \in \{0^\circ, 5^\circ, \dots, 45^\circ\}$ we compute $F_\theta(r) = |X_\theta \circ D_r|/|X_\theta|$, implementing openings by chain elements as cascades of erosions and dilations by their elementary sets. We report the rotation sensitivity $S = \text{mean}_r \text{std}_\theta F_\theta(r)$ and the number of increases $F_\theta(r+1) > F_\theta(r)$, which are impossible for a granulometry. The scenes are nine squares of sides 17 to 81 pixels, rasterised exactly after rotation ($r \leq 40$), and three real images from `scikit-image`: fluorescence microscopy of nuclei (`human_mitosis`, $r \leq 15$), `coins` ($r \leq 30$) and a material texture (`gravel`, $r \leq 20$). The real images are smoothed, rotated with bilinear interpolation and thresholded at the Otsu level of the unrotated image.

Table 4 and Fig. 2 illustrate the dilemma of the title. On the synthetic scene, Gauss disks are the least orientation dependent but produce 17 increases, i.e. negative pattern-spectrum values. Octagons are correct but strongly orientation dependent, and the periodic-line chain divides their sensitivity by three. On the three real images, the families differ by at most $0.6 \cdot 10^{-3}$ (nuclei, gravel) and

Table 4. Rotation sensitivity $S (\times 10^3)$ of size distributions; in parentheses, the number of increases of F_θ when nonzero.

scene	Gauss	octagons	DSS 8 dir.	periodic 8	periodic 13	DSS MILP
squares (synthetic)	4.0 (17)	30.2	13.3	11.8	9.5	17.6
nuclei (microscopy)	1.7	1.9	1.9	2.0	1.6	1.4
coins	5.0	5.0	6.4	4.2	5.0	6.2
gravel	1.4	1.3	1.6	1.1	1.6	1.2

**Fig. 2.** Size distributions of the rotated synthetic scene, one curve per angle: Gauss disks are nearly isotropic but non-monotone; octagons are monotone but orientation dependent; the periodic-line chain is monotone and three times less orientation dependent.

$2.2 \cdot 10^{-3}$ (coins), with no consistent ranking, i.e. within the effect of interpolation and thresholding; no increases occur there. Rounded, randomly oriented objects average out the anisotropy of the structuring element. The choice of family therefore matters most for polygonal or aligned structures. The violation of the axiom by Gauss disks, in contrast, is a correctness issue that can surface on any image (Table 1).

7 Conclusion

Standard digital disks do not generate granulometries, and granulometric Minkowski chains cannot be uniformly round. With mean-radius steps δ their error is $\Theta(\rho/\delta^2)$, and $\Theta(\sqrt{\rho})$ when $\delta \sim \rho^{1/4}$; bounded steps force linear growth in every dimension. For practice, one must choose between correct but anisotropic chains, whose loss the constant κ quantifies, and round but incorrect digital disks. Our periodic-line chains halve the loss of octagons. Three questions remain open: the optimal rate when only the fitted radius has bounded steps, between $\rho^{1/3}$ and $\rho^{1/2}$; whether granulometries that are not Minkowski chains, where D_l is merely a union of translates of D_k , can escape the bound; and matching upper bounds in $\mathbb{Z}^3$.

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